

Jay Knight

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EDUCATION **University of Nevada, Reno**
B.S. Computer Science and Engineering | *Minor in Mathematics and Robotics* *Expected 2026*
M.S. Robotics *Expected 2027*

COURSEWORK Algorithms, Data Structures, Systems Administration, Embedded Systems, Motion Planning, Mobile Robots, Socially Assistive Robotics

EXPERIENCE **Undergraduate Researcher | University of Nevada, Reno** *Aug 2023 - Present*

- Built modular ROS 2 software for customizable social-robot faces, enabling dynamic behavior configuration for AI Institute for Exceptional Education studies.
- Developed a full-stack web app with Flask and MujocoSim, integrating simulation and real-time feedback to accelerate learning robotics concepts using AI assistants.
- Implemented robot drivers in C++ and Python on embedded Linux platforms, reducing per-unit hardware costs for community-college robotics kits.

REU Undergraduate Researcher | University of Nevada, Reno *Jun 2023 - Aug 2023*

- Designed and tested methods for generally applicable and architectural solutions for human-robot joint tasks.
- Evaluated performance in kitchen-task scenarios, informing ongoing joint-task autonomy research.

K-12 Robotics Education Assistant | Nevada Robotics | Reno, NV *Nov 2021 - Aug 2024*

- Managed statewide robotics lending library and trained 500+ teachers across Nevada.
- Modernized robotics lending system, and created integrated reporting pipelines for grant proposals.

PUBLICATIONS *Design Activity for Personalized Robot Faces: Evaluating Child Responses To Customizable Expressive Faces.* IEEE/ACM RO-MAN 2025.
Architectural Solution for Tightly-Coupled Robot Collaborative Tasks. Submitted. International Conference on Intelligent Robots and Systems 2024.

PROJECTS **HomeLab** | Providing streaming services for personal use and low-latency remote desktop streaming to virtual machines for remote development, and system administration tasks.
Robot Restoration | Ongoing diagnosis, maintenance, and repair of robots in the UNR Robotics Center, including *PR2*, *Quori*, *Misty II*, and *Blossom*.
'RPI Robot' | Designed and built differential-drive robot with LiDAR, encoders, and ROS 2 navigation; used in community-college robotics curriculum.

LEADERSHIP **UNR Robotics Club** | President, Co-Founder *Dec 2022 - May 2024*

- Secured \$10k in robotics education and team development grants
- Created a robotics community resource to Reno K-12 competition teams

FBLA Nevada State Champion | Mobile Application Development *May 2018*

- Collaborated with team members to develop a social media application for recycling second hand belongings.

SKILLS Linux, MacOS, Windows | C/C++, Python, ROS, .NET Core, NodeJS, Bash, Git, Wordpress | AutoCAD, Blender | Research, Web Development, Azure Deployments, Systems Administration, Project Management, Interdisciplinary Collaboration, Technical Writing, Presentation